

EX3: Guided Microwave Propagation

The purpose of this lab is to develop experimental techniques to generate microwaves and observe how they propagate in rectangular waveguides. Microwaves typically operate in the 2.45 GHz frequency range.

1. Finding ratio η

For any electromagnetic wave in free space, we know that we can relate the E field to the magnetic flux density B field by c (speed of light) and the H field by μ_0 (permeability of free space):

$$E = cB \quad (1.1)$$

$$H = \frac{B}{\mu_0} \quad (1.2)$$

where $c = 3 \times 10^8$ m/s and $\mu_0 = 4\pi \times 10^{-7}$ N/A².

Plugging this back into the formula for η , we get:

$$\begin{aligned} \eta &= \frac{E}{H} \\ \eta &= \frac{cB}{\frac{B}{\mu_0}} \\ &= c\mu_0 \\ &= \boxed{377\Omega} \end{aligned} \quad (1.3)$$

2. E-field reflection coefficient

At the interface between two regions, we know there are 3 different E-fields:

$$E_i = E_o \quad (2.1)$$

$$E_r = \rho E_o \quad (2.2)$$

$$E_t = \tau E_o \quad (2.3)$$

where E_i = incident field, E_r = reflected field, E_t = transmitted field.

From here, we can apply boundary conditions at the interface to find the reflection ρ and transmission τ coefficients. We assume the incident plane wave is striking the interface at 90 deg, so the E-fields have no normal components.

a. Continuity of tangential electric field:

$$\begin{aligned}
 E_i + E_r &= E_t \\
 E_o + \rho E_o &= \tau E_o \\
 1 + \rho &= \tau
 \end{aligned}
 \tag{2.4}$$

b. Continuity of tangential magnetic field (note: if $\eta_1 < \eta_2$, the reflected H-field will undergo a 180 deg phase shift):

$$\begin{aligned}
 H_i - H_r &= H_t \\
 \frac{E_i}{\eta_1} - \frac{E_r}{\eta_1} &= \frac{E_t}{\eta_2} \\
 \frac{E_o}{\eta_1} - \frac{\rho E_o}{\eta_1} &= \frac{\tau E_o}{\eta_2} \\
 \frac{1 - \rho}{\eta_1} &= \frac{\tau}{\eta_2}
 \end{aligned}
 \tag{2.5}$$

Solving this system of two equations (2.4 and 2.5), we get an expression for both ρ and τ :

$$\begin{aligned}
 \frac{1 - \rho}{\eta_1} &= \frac{1 + \rho}{\eta_2} \\
 \eta_2 - \eta_2 \rho &= \eta_1 + \eta_1 \rho \\
 \rho &= \boxed{\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}}
 \end{aligned}
 \tag{2.6}$$

$$\begin{aligned}
 \frac{1 - (\tau - 1)}{\eta_1} &= \frac{\tau}{\eta_2} \\
 2\eta_2 - \tau\eta_2 &= \tau\eta_1 \\
 \tau &= \boxed{\frac{2\eta_2}{\eta_1 + \eta_2}}
 \end{aligned}
 \tag{2.7}$$

3. Plotting Smith charts

What is it? Smith chart is a graphing tool used in radio freq engineering to "analyze and design impedance-matching circuits, transmission lines, and antennas" - Google. It's a polar plot that visualizes/maps reflection and transmission coefficients.

```
In [84]: import numpy as np
import matplotlib.pyplot as plt
```

3.1 Fixed resistance

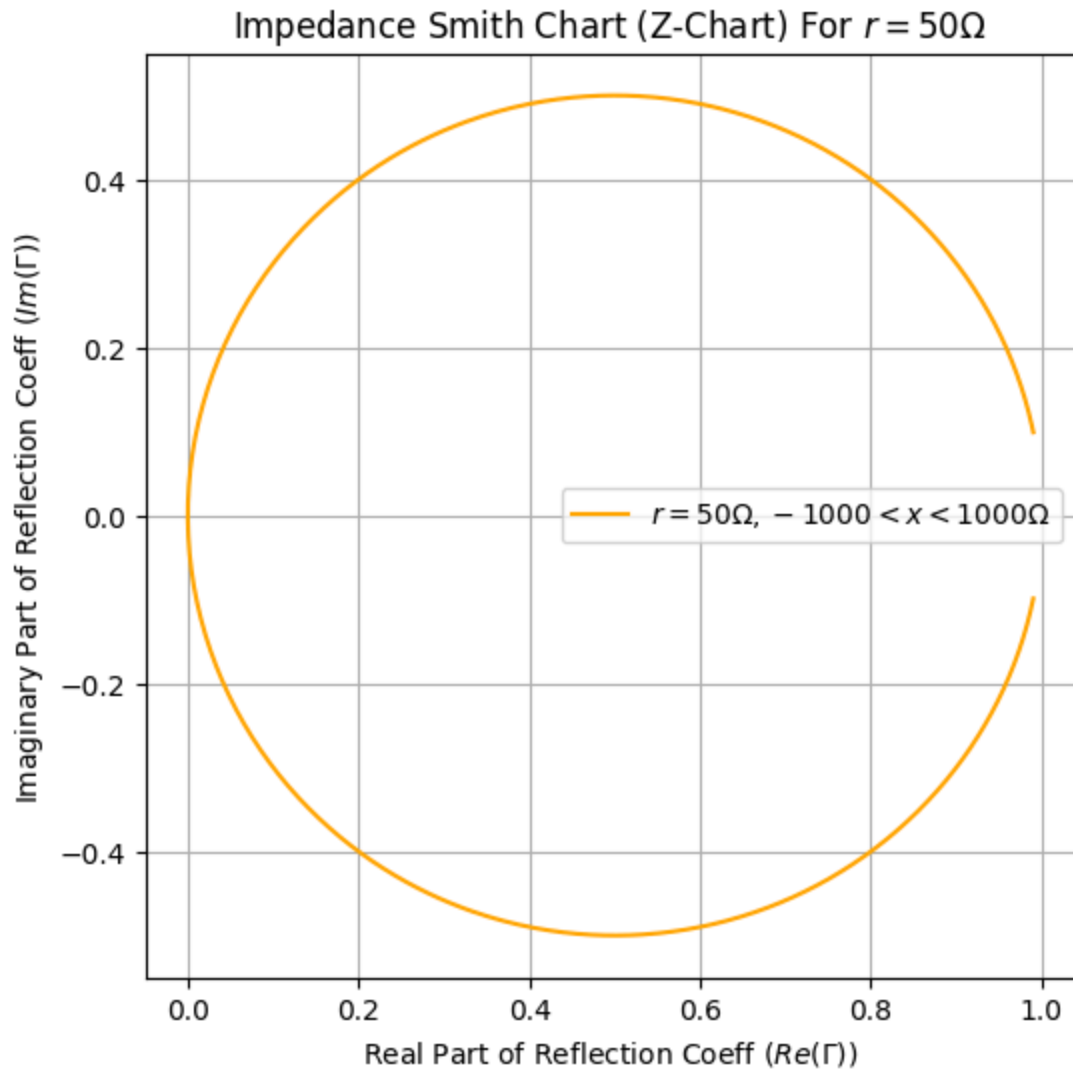
```
In [85]: z_0 = 50          # Characteristic waveguide impedance
r = 50      # Re(z) where z = genral complex impedance
x_min = -1000
x_max = 1000
```

```

z = r + 1j * np.linspace(x_min, x_max, 2000)
gamma = (z - z_0) / (z + z_0)      # Complex reflection coefficients

plt.figure(figsize=(6, 6))
plt.plot(gamma.real, gamma.imag, label=r"$r = 50 \Omega, -1000 < x < 1000 \Omega$")
plt.xlabel(r"Real Part of Reflection Coeff ($Re(\Gamma)$)")
plt.ylabel(r"Imaginary Part of Reflection Coeff ($Im(\Gamma)$)")
plt.title(r"Impedance Smith Chart (Z-Chart) For $r = 50 \Omega$")
plt.legend()
plt.grid()
plt.show()

```



This is what the Smith chart looks like for $r = 50\Omega$. Some observations:

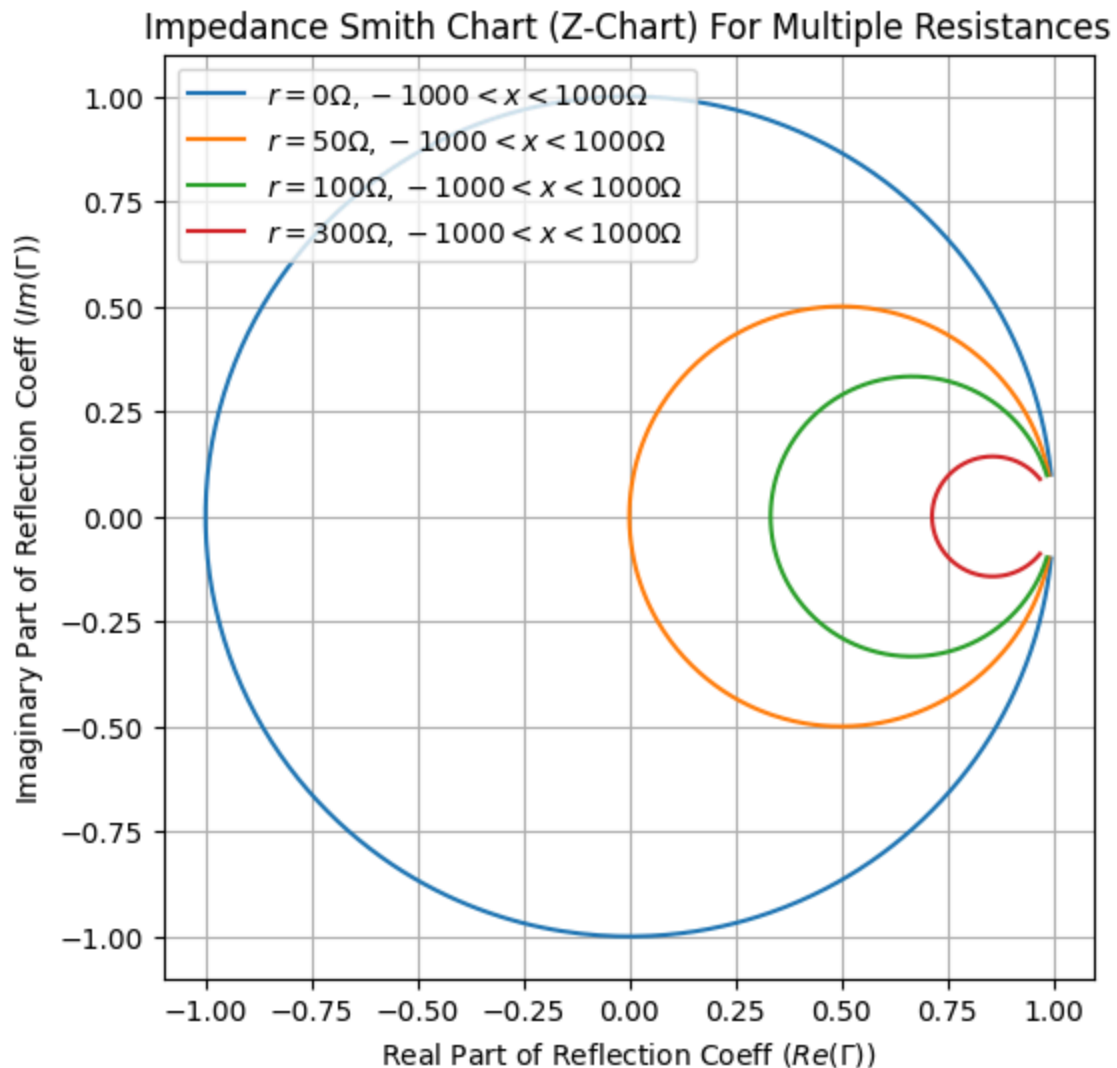
- Plot forms a perfect circle that's nearly complete - missing segment occurs around $(1, 0)$
- Plot passes through origin ($\Gamma = 0$) when $z = z_0$

3.2 Multiple resistances

```
In [86]: def smith_chart_multiple_r():
    z_0 = 50 # Characteristic waveguide impedance
    r = [0, 50, 100, 300] # Resistance
    x_min = -1000
    x_max = 1000

    for r_i in r:
        z = r_i + 1j * np.linspace(x_min, x_max, 2000)
        gamma = (z - z_0) / (z + z_0)
        plt.plot(gamma.real, gamma.imag, label=rf"$r = {r_i} \Omega, -1000 < x < 1000 \Omega$")

    plt.figure(figsize=(6, 6))
    smith_chart_multiple_r()
    plt.xlabel(r"Real Part of Reflection Coeff ($Re(\Gamma)$)")
    plt.ylabel(r"Imaginary Part of Reflection Coeff ($Im(\Gamma)$)")
    plt.title(r"Impedance Smith Chart (Z-Chart) For Multiple Resistances")
    plt.legend()
    plt.grid()
    plt.show()
```



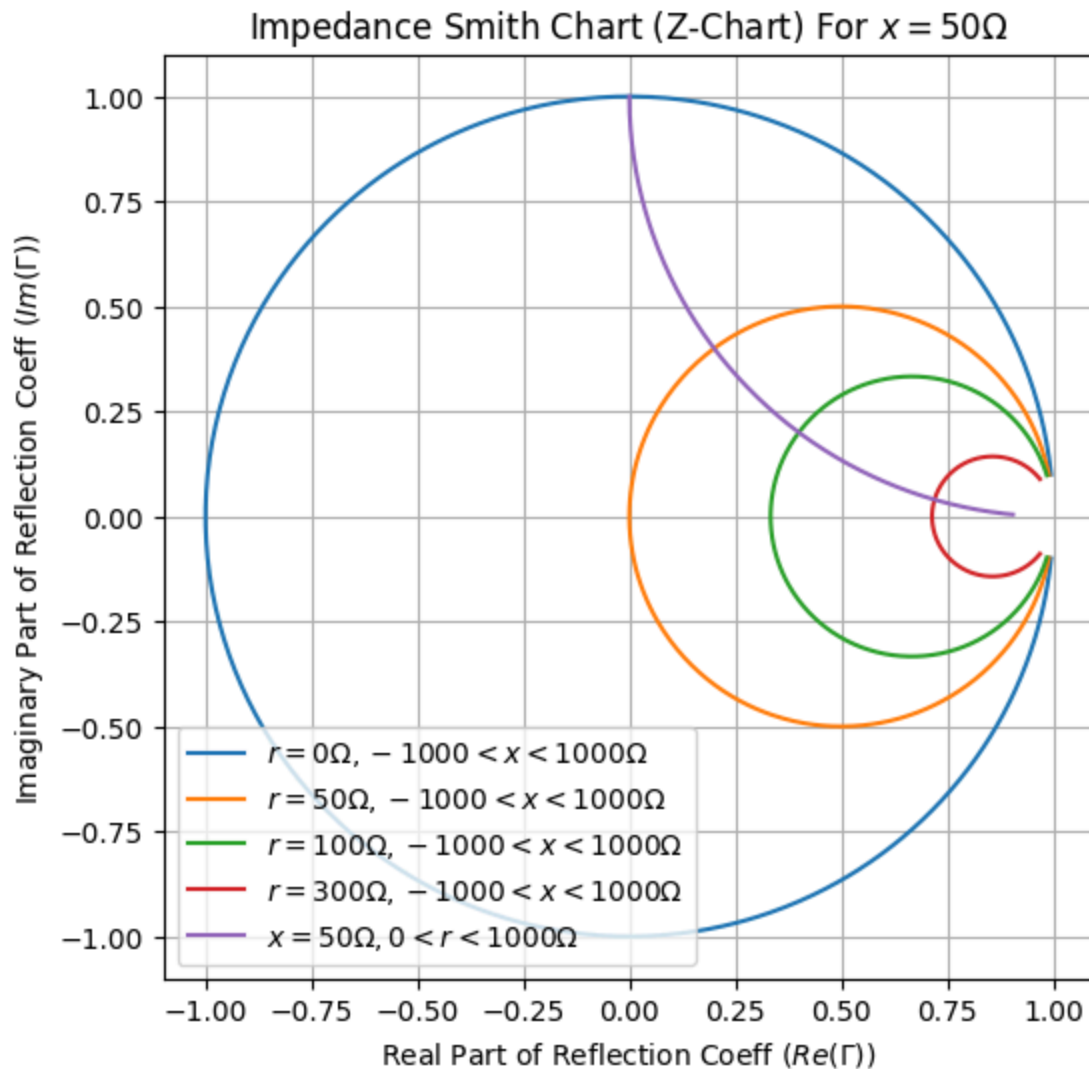
This is what the Smith chart looks like for multiple r values. Some observations:

- Circles get smaller as r increases with centers moving further to the right
- $r = 0\Omega$ gives us a unit circle centered around the origin
- The unit circle here means $|\Gamma| = 1$, which means we get input fields are 100% reflected
- ^ This makes sense since $r = 0$ which means no energy is dissipated as heat, load is likely made up of only inductors and capacitors

3.3 Fixed reactance

```
In [87]: def smith_chart_fixed_x():
    x = 50
    z = np.linspace(0, 1000, 1000) + 1j * x
    gamma = (z - z_0) / (z + z_0)
    plt.plot(gamma.real, gamma.imag, label=rf"$x = 50 \backslash\Omega, 0 < r < 1000 \backslash\Omega$")

plt.figure(figsize=(6, 6))
smith_chart_multiple_r()
smith_chart_fixed_x()
plt.xlabel(r"Real Part of Reflection Coeff ($Re(\backslash\Gamma)$)")
plt.ylabel(r"Imaginary Part of Reflection Coeff ($Im(\backslash\Gamma)$)")
plt.title(r"Impedance Smith Chart (Z-Chart) For $x = 50 \backslash\Omega$")
plt.legend()
plt.grid()
plt.show()
```



This is what the Smith chart looks like for $x = 50\Omega$ (purple line). Some observations:

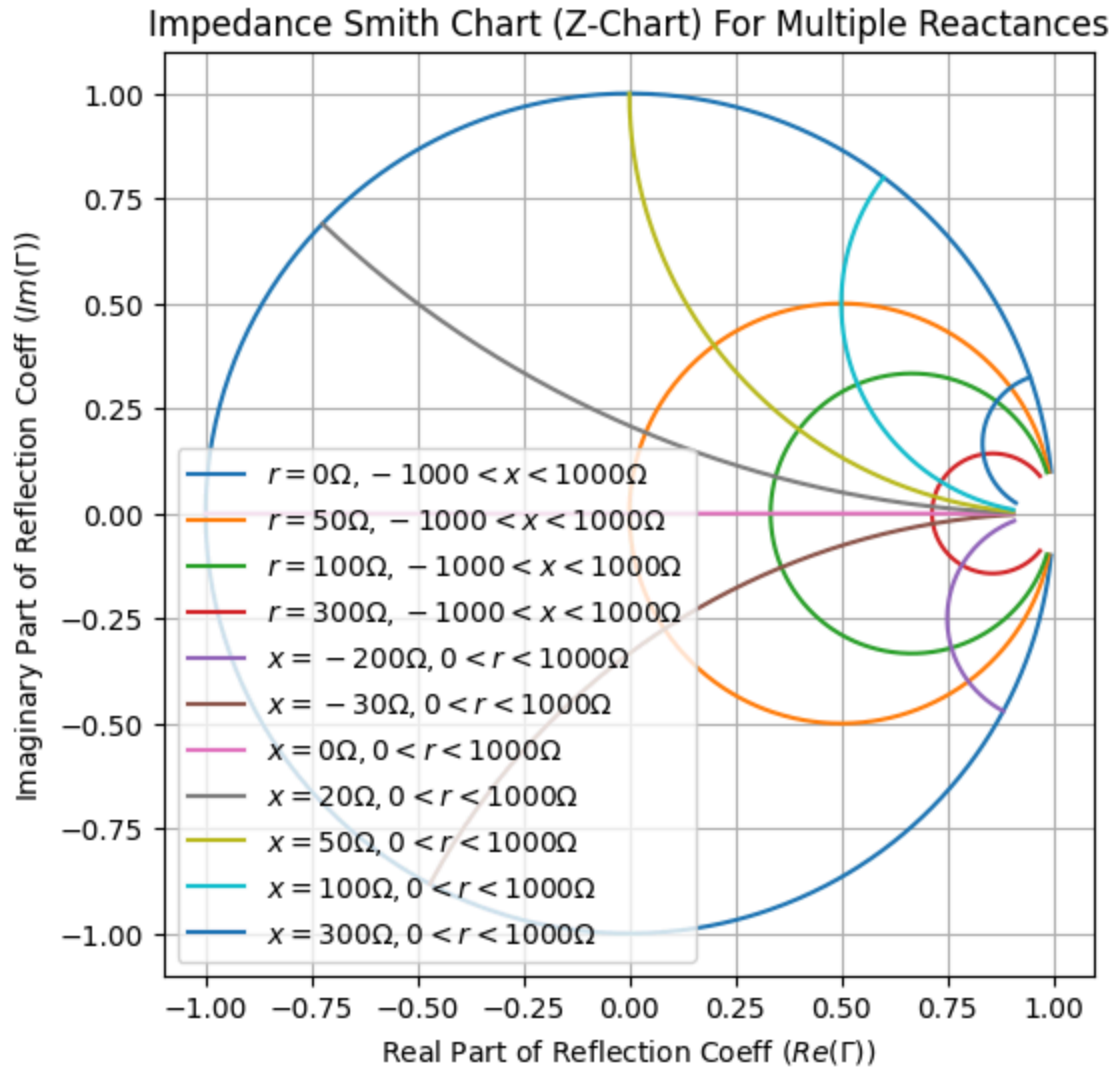
- The purple curve isn't concentric like the other circles
- In fact, it actually intersects the fixed resistance circles at 90° angles

3.4 Multiple reactances

```
In [88]: def smith_chart_multiple_x():
x = [-200, -30, 0, 20, 50, 100, 300]
for x_i in x:
    z = np.linspace(0, 1000, 1500) + 1j * x_i
    gamma = (z - z_0) / (z + z_0)
    plt.plot(gamma.real, gamma.imag, label=rf"$x = {x_i} \Omega, 0 < r < 1000\Omega$")

plt.figure(figsize=(6, 6))
smith_chart_multiple_r()
smith_chart_multiple_x()
plt.xlabel(r"Real Part of Reflection Coeff ( $Re(\Gamma)$ )")
plt.ylabel(r"Imaginary Part of Reflection Coeff ( $Im(\Gamma)$ )")
plt.title(r"Impedance Smith Chart (Z-Chart) For Multiple Reactances")
plt.legend()
```

```
plt.grid()
plt.show()
```



This is what the Smith chart looks like for multiple x values. Some observations:

- These new fixed x curves are only part-circles since we don't let r drop below 0 (we assume no negative resistances)
- Seem like positive x values create inductive curves in the top half of the imaginary plane since V_L leads V_o
- ^ Negative x values form capacitive curves in bottom half since V_c lags V_o